Pain Knowledge Tree — Final Mentor-Submission Version

Project: Smart Body Health Monitoring System Using Contactless Body Scanning

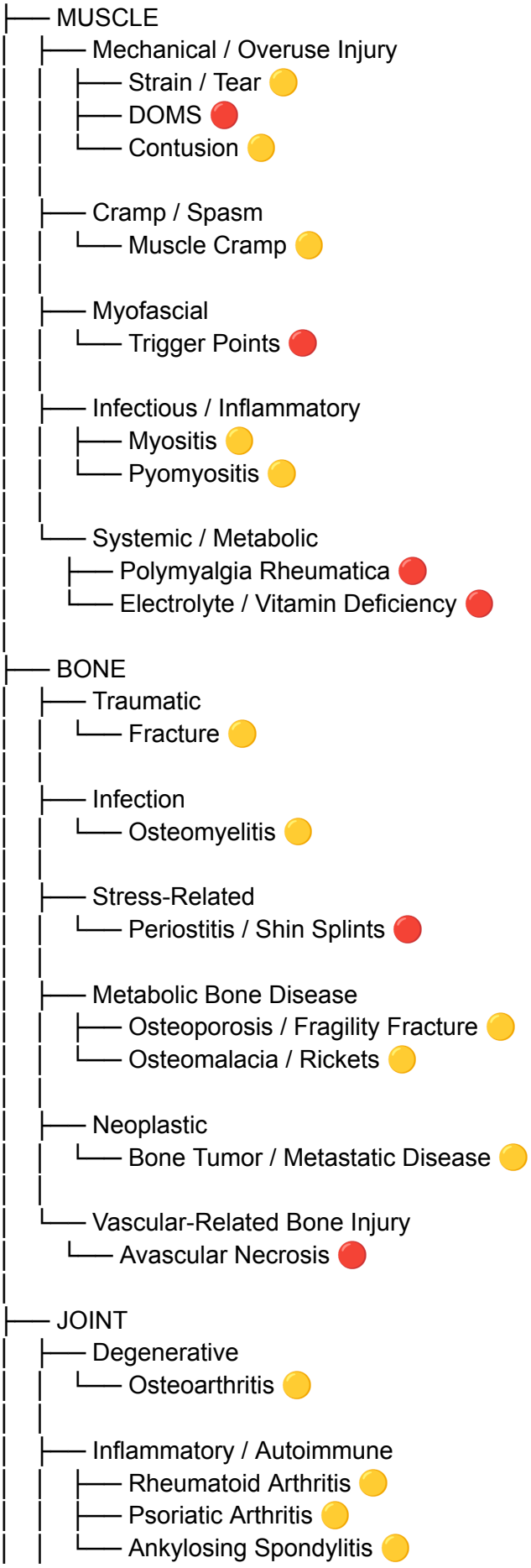
Source: Pain_Knowledge_Tree_Refined.md — reformatted for Google Docs compatibility. No scientific content changed; see Section 6 for what was fixed and why.

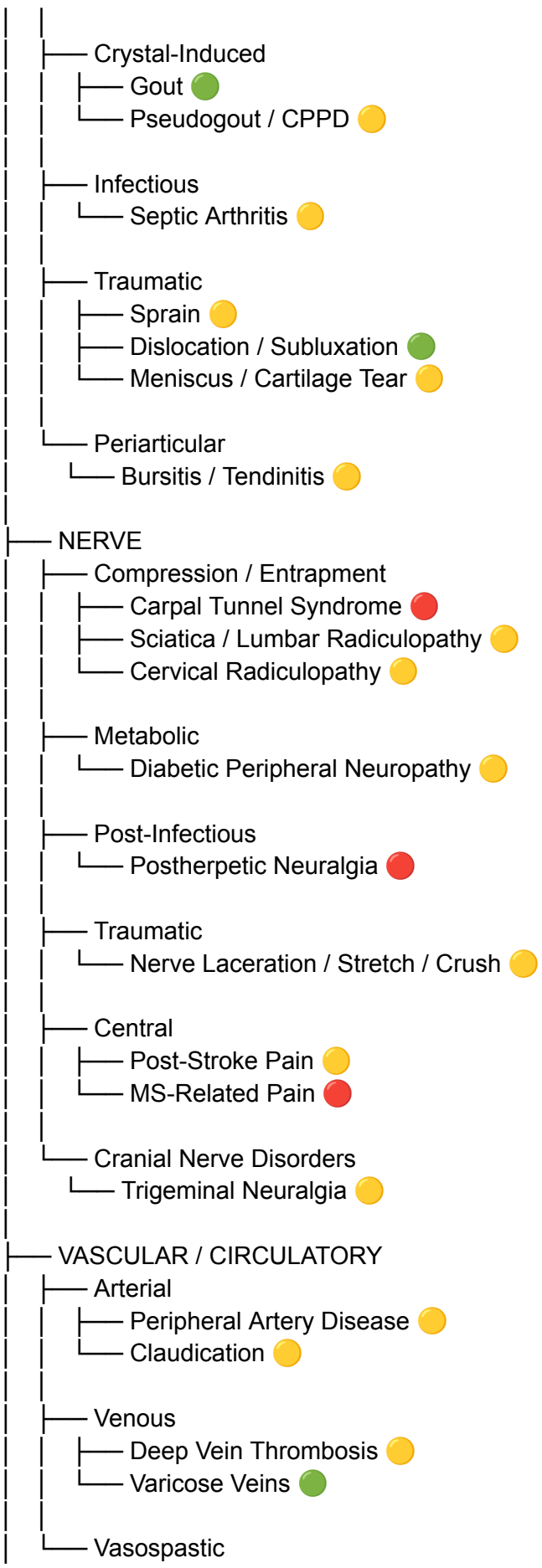
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1. Final Master Tree

(Complete tree — all tissues, causes, and conditions, with detectability icon. Mechanism, attributes, and contactless method are kept in separate sections below rather than inline, so the tree stays short enough to convert into boxes.)

PAIN
│
├── SKIN
│ ├── Mechanical Injury
│ │ ├── Cut / Laceration 🟡
│ │ ├── Abrasion 🟢
│ │ ├── Puncture Wound 🟡
│ │ └── Contusion / Bruise 🟢
│ │
│ ├── Burn
│ │ ├── Thermal Burn 🟡
│ │ ├── Chemical Burn 🟡
│ │ └── Sunburn 🟢
│ │
│ ├── Infection
│ │ ├── Cellulitis / Abscess 🟡
│ │ ├── Shingles 🟡
│ │ └── Fungal Infection 🟡
│ │
│ ├── Inflammatory / Allergic
│ │ ├── Contact Dermatitis 🟢
│ │ └── Eczema / Psoriasis Flare 🟢
│ │
│ └── Pressure / Ischemic Injury
│ └── Pressure Ulcer 🟢
│


MUSCLE
Mechanical / Overuse Injury
Strain / Tear 🟡
DOMS 🔴
Contusion 🟡
Cramp / Spasm
Muscle Cramp 🟡
Myofascial
Trigger Points 🔴
Infectious / Inflammatory
Myositis 🟡
Pyomyositis 🟡
Systemic / Metabolic
Polymyalgia Rheumatica 🔴
Electrolyte / Vitamin Deficiency 🔴
BONE
Traumatic
Fracture 🟡
Infection
Osteomyelitis 🟡
Stress-Related
Periostitis / Shin Splints 🔴
Metabolic Bone Disease
Osteoporosis / Fragility Fracture 🟡
Osteomalacia / Rickets 🟡
Neoplastic
Bone Tumor / Metastatic Disease 🟡
Vascular-Related Bone Injury
Avascular Necrosis 🔴
JOINT
Degenerative
Osteoarthritis 🟡
Inflammatory / Autoimmune
Rheumatoid Arthritis 🟡
Psoriatic Arthritis 🟡
Ankylosing Spondylitis 🟡


Crystal-Induced
Gout 🟢
Pseudogout / CPPD 🟡
Infectious
Septic Arthritis 🟡
Traumatic
Sprain 🟡
Dislocation / Subluxation 🟢
Meniscus / Cartilage Tear 🟡
Periarticular
Bursitis / Tendinitis 🟡
NERVE
Compression / Entrapment
Carpal Tunnel Syndrome 🔴
Sciatica / Lumbar Radiculopathy 🟡
Cervical Radiculopathy 🟡
Metabolic
Diabetic Peripheral Neuropathy 🟡
Post-Infectious
Postherpetic Neuralgia 🔴
Traumatic
Nerve Laceration / Stretch / Crush 🟡
Central
Post-Stroke Pain 🟡
MS-Related Pain 🔴
Cranial Nerve Disorders
Trigeminal Neuralgia 🟡
VASCULAR / CIRCULATORY
Arterial
Peripheral Artery Disease 🟡
Claudication 🟡
Venous
Deep Vein Thrombosis 🟡
Varicose Veins 🟢
Vasospastic

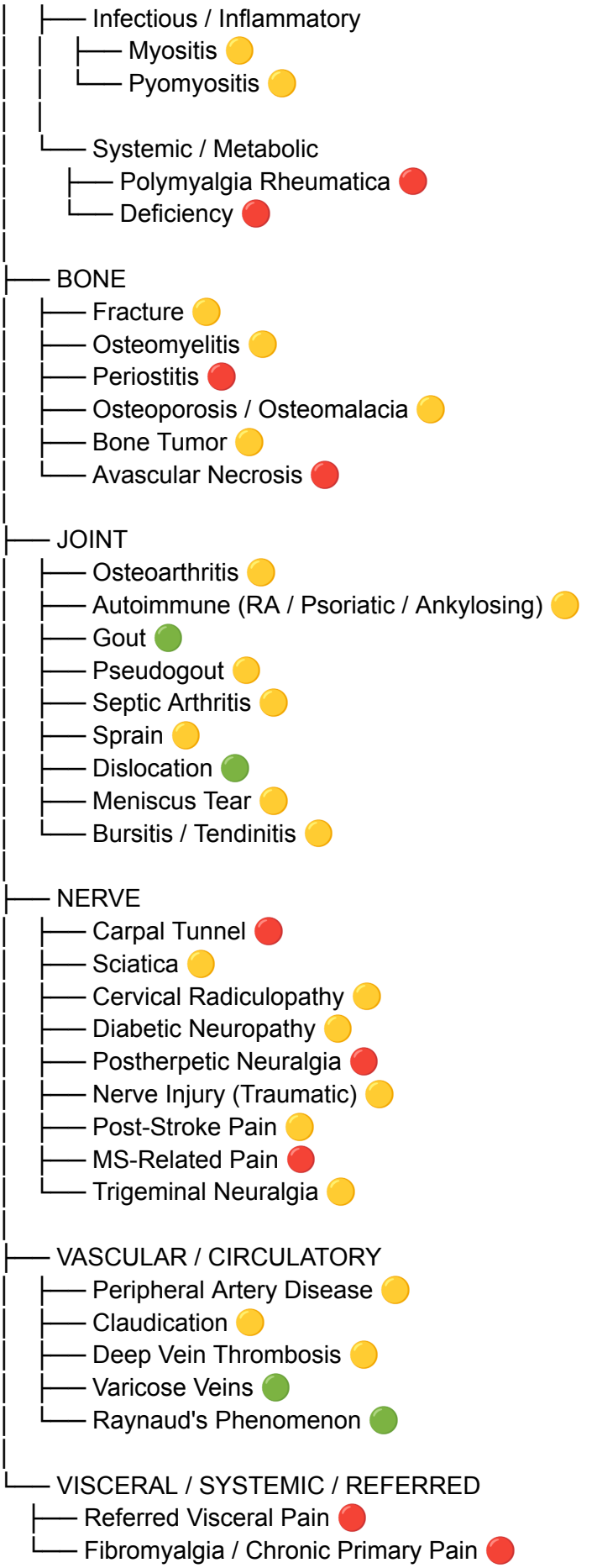
```
│       └── Raynaud's Phenomenon 🟢
│
└── VISCERAL / SYSTEMIC / REFERRED
    ├── Referred Visceral Pain 🔴
    └── Chronic Primary / Nociplastic Pain 🔴
        └── includes Fibromyalgia (single node)
```

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2. Final Mentor-Submission Tree

(Same hierarchy, stripped to the minimum — condition name and icon only. This is the version to convert into boxes and arrows in Google Drawings.)

```
PAIN
│
├── SKIN
│   ├── Mechanical Injury
│   │   ├── Cut / Laceration 🟡
│   │   ├── Abrasion 🟢
│   │   └── Bruise 🟢
│   │
│   ├── Burn
│   │   ├── Thermal Burn 🟡
│   │   └── Sunburn 🟢
│   │
│   ├── Infection
│   │   ├── Cellulitis 🟡
│   │   ├── Shingles 🟡
│   │   └── Fungal Infection 🟡
│   │
│   ├── Inflammatory / Allergic
│   │   ├── Contact Dermatitis 🟢
│   │   └── Eczema / Psoriasis 🟢
│   │
│   └── Pressure Injury
│       └── Pressure Ulcer 🟢
│
├── MUSCLE
│   ├── Overuse Injury
│   │   ├── Strain / Tear 🟡
│   │   ├── DOMS 🔴
│   │   └── Contusion 🟡
│   │
│   ├── Cramp / Spasm 🟡
│   ├── Myofascial (Trigger Points) 🔴
│   │
```


Infectious / Inflammatory
Myositis 🟡
Pyomyositis 🟡
Systemic / Metabolic
Polymyalgia Rheumatica 🔴
Deficiency 🔴
BONE
Fracture 🟡
Osteomyelitis 🟡
Periostitis 🔴
Osteoporosis / Osteomalacia 🟡
Bone Tumor 🟡
Avascular Necrosis 🔴
JOINT
Osteoarthritis 🟡
Autoimmune (RA / Psoriatic / Ankylosing) 🟡
Gout 🟢
Pseudogout 🟡
Septic Arthritis 🟡
Sprain 🟡
Dislocation 🟢
Meniscus Tear 🟡
Bursitis / Tendinitis 🟡
NERVE
Carpal Tunnel 🔴
Sciatica 🟡
Cervical Radiculopathy 🟡
Diabetic Neuropathy 🟡
Postherpetic Neuralgia 🔴
Nerve Injury (Traumatic) 🟡
Post-Stroke Pain 🟡
MS-Related Pain 🔴
Trigeminal Neuralgia 🟡
VASCULAR / CIRCULATORY
Peripheral Artery Disease 🟡
Claudication 🟡
Deep Vein Thrombosis 🟡
Varicose Veins 🟢
Raynaud's Phenomenon 🟢
VISCERAL / SYSTEMIC / REFERRED
Referred Visceral Pain 🔴
Fibromyalgia / Chronic Primary Pain 🔴

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3. Pain Mechanism Supporting Tree

(Separate concept — how pain is generated/perceived, not where it originates)

PAIN MECHANISMS
├── Nociceptive — tissue damage (most conditions above)
├── Neuropathic — nerve damage/dysfunction (e.g. Sciatica, Postherpetic Neuralgia, Trigeminal Neuralgia)
└── Nociplastic — altered pain processing, no clear damage (e.g. Fibromyalgia)

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4. Supporting Attribute Information

(Grouped by tissue. Each table has only 4 short columns, so it pastes cleanly into Google Docs.)

SKIN

Condition	Observable	Non-Observable	Detectability
Cut / Laceration	Wound, bleeding	Depth	🟡
Abrasion	Scraped skin, redness	Debris	🟢
Puncture Wound	Entry wound, swelling	Depth	🟡
Bruise	Colour change	Deep extent	🟢
Thermal Burn	Redness, blistering	True depth	🟡
Chemical Burn	Discoloration	Chemical ID	🟡
Sunburn	Erythema, swelling	DNA injury	🟢
Cellulitis	Redness, warmth	Organism	🟡
Shingles	Dermatomal rash	Pre-rash pain	🟡
Fungal Infection	Scaling	Invasion depth	🟡
Contact Dermatitis	Bordered rash	Allergen	🟢

Eczema / Psoriasis	Plaques, scaling	Disease activity	🟢
Pressure Ulcer	Discoloration	Necrosis depth	🟢

Typical contactless method: RGB camera; thermal camera adds warmth signal for infections/burns.

MUSCLE

Condition	Observable	Non-Observable	Detectability
Strain / Tear	Swelling, guarded gait	Tear grade	🟡
DOMS	Usually none	Micro-damage	🔴
Contusion	Swelling, discoloration	Depth	🟡
Muscle Cramp	Bulging (episode only)	Trigger cause	🟡
Trigger Points	Usually none	Taut band	🔴
Myositis	Skin signs (some types)	Biopsy/enzymes	🟡
Pyomyositis	Swelling if superficial	Abscess depth	🟡
Polymyalgia Rheumatica	Reduced movement	Inflammatory markers	🔴
Electrolyte Deficiency	None specific	Serum levels	🔴

Typical contactless method: RGB camera + pose estimation for gait/guarding.

BONE

Condition	Observable	Non-Observable	Detectability
Fracture	Deformity, swelling	Fracture pattern	🟡
Osteomyelitis	Swelling if superficial	Marrow extent	🟡
Periostitis	Local tenderness	Stress-fracture risk	🔴
Osteoporosis	Postural change (late)	Bone density	🟡
Osteomalacia	Deformity (severe)	Vitamin D status	🟡

Bone Tumor	Mass (some cases)	Tumor type/stage	🟡
Avascular Necrosis	Usually none early	Ischemic extent	🔴

Typical contactless method: RGB + pose estimation; not a substitute for X-ray.

JOINT

Condition	Observable	Non-Observable	Detectability
Osteoarthritis	Swelling, altered gait	Cartilage loss	🟡
Rheumatoid Arthritis	Deformity (advanced)	Autoantibodies	🟡
Psoriatic Arthritis	Swelling, skin plaques	Joint damage	🟡
Ankylosing Spondylitis	Posture change (advanced)	Sacroiliac status	🟡
Gout	Marked redness, warmth	Crystal type	🟢
Pseudogout	Redness, warmth	Crystal type	🟡
Septic Arthritis	Marked swelling, warmth	Organism	🟡
Sprain	Swelling, instability	Ligament grade	🟡
Dislocation	Obvious deformity	Nerve/vessel injury	🟢
Meniscus Tear	Swelling, altered gait	Tear pattern	🟡
Bursitis / Tendinitis	Localized swelling	Inflammation extent	🟡

Typical contactless method: RGB + gait/pose analysis; thermal adds warmth signal.

NERVE

Condition	Observable	Non-Observable	Detectability
Carpal Tunnel	Wasting (advanced only)	Nerve function	🔴
Sciatica	Antalgic gait/posture	Disc pathology	🟡
Cervical Radiculopathy	Posture compensation	Nerve root status	🟡
Diabetic Neuropathy	Ulcers (advanced)	Nerve function	🟡

Postherpetic Neuralgia	Residual scarring	Nerve damage extent	🔴
Nerve Injury (Traumatic)	Wasting, positioning	Damage extent	🟡
Post-Stroke Pain	Weakness, gait asymmetry	Lesion location	🟡
MS-Related Pain	Gait changes (may develop)	Lesion type	🔴
Trigeminal Neuralgia	Grimacing (attack only)	Compression cause	🟡

Typical contactless method: pose/gait analysis; most nerve pathology itself is not visible.

VASCULAR / CIRCULATORY

Condition	Observable	Non-Observable	Detectability
Peripheral Artery Disease	Pallor, gait change	Obstruction degree	🟡
Claudication	Limping, temp change	Narrowing degree	🟡
Deep Vein Thrombosis	Swelling, warmth	Clot extent	🟡
Varicose Veins	Dilated, bulging veins	Deep vein function	🟢
Raynaud's Phenomenon	Finger/toe colour change	Secondary cause	🟢

Typical contactless method: RGB + thermal camera.

VISCERAL / SYSTEMIC / REFERRED

Condition	Observable	Non-Observable	Detectability
Referred Visceral Pain	Sweating, pallor	Organ pathology	🔴
Fibromyalgia	No reliable signature	Symptom pattern	🔴

No reliable contactless method for this branch; clinical evaluation required.

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5. Detectability Legend

Icon	Label	Meaning
🟢	YES	A relatively strong observable feature may be detectable through a contactless method.
🟡	PARTIAL	An associated feature may be observable, but it is non-specific and cannot alone establish cause or severity.
🔴	NO	No reliable camera-only signature should be assumed.

Note: 🟢 never means "the camera can diagnose this." It means an associated observable feature (redness, swelling, discoloration, posture/gait change, temperature asymmetry) may be detected — never a diagnosis.

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6. Final Refinements Made

(Changes made in this formatting pass, from the previous mentor-submission draft)

Change 1 — Replaced the single wide 8-column table with per-tissue 4-column tables

- *Original:* One long table (Tissue, Major Cause, Condition, Mechanism, Observable, Non-Observable, Method, Detectability) with full-sentence cells.
- *Revised:* Seven short tables (one per tissue), each with exactly 4 columns (Condition, Observable, Non-Observable, Detectability) and 1–3 word cell entries. Contactless method moved to one summary line under each table instead of a table column.
- *Reason:* Google Docs renders wide tables and long cell text as extremely narrow columns with one character per line. Splitting by tissue and cutting columns to 4 with short entries avoids this.

Change 2 — Removed inline observable/non-observable text from both trees

- *Original:* Master Tree carried parenthetical descriptions (e.g., "open wound, bleeding, edges") on every node.
- *Revised:* Both trees now show only Condition + Detectability icon; attribute detail lives solely in Section 4.
- *Reason:* Long inline text breaks node shapes when converted to boxes in Google Drawings; keeping nodes to a few words keeps them shape-ready.

Change 3 — Removed mechanism tags from tree nodes

- *Original:* Nodes carried a `[Mechanism]` tag alongside the icon.
- *Revised:* Mechanism is shown only in the separate Pain Mechanism Supporting Tree (Section 3).

- *Reason:* Keeps each node to "Condition + icon" only, per the requested format, while still preserving the mechanism concept as its own layer.

No condition, cause category, tissue branch, or detectability rating was added, removed, or reclassified — only presentation changed.

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7. Potential Contactless Monitoring Areas

(Branch-level only — not a final ML scope)

A. Strong candidates (majority 🟢)

- Skin — mechanical injury, burns, inflammatory/allergic conditions, pressure injury
- Joint — crystal-induced and traumatic-deformity conditions
- Vascular — venous and vasospastic conditions

B. Partial candidates (🟡 majority — screening flag only)

- Joint — degenerative and autoimmune conditions
- Bone — traumatic and infectious conditions
- Muscle — mechanical/overuse and infectious/inflammatory conditions
- Nerve — compression and central conditions (gait/posture proxies only)
- Vascular — arterial conditions

C. Poor candidates (🔴 — no reliable visual signature)

- Muscle — myofascial and systemic/metabolic conditions
- Bone — stress-related and vascular-related conditions
- Nerve — post-infectious and select central conditions
- Visceral/Systemic — referred and chronic primary/nociplastic pain

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8. Project Scope Boundary

This project does **not** claim that contactless scanning diagnoses pain.

The accurate framing is: *contactless scanning may identify observable physical indicators associated with certain painful conditions and provide a preliminary monitoring/screening indication.*

Deep tissue, neurological, vascular, and systemic/visceral causes may require clinical examination, laboratory testing, or medical imaging that a camera-based system cannot replace.

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9. Short Conclusion

This tree organizes pain by where it originates (tissue/source), why it occurs (cause/condition), and what is observable from outside the body — kept separate from how the pain is generated (mechanism) and from what a contactless sensor could plausibly detect (detectability). Formatting is now Google Docs–safe: short tree nodes and small per-tissue tables instead of one wide table. Pending mentor approval, the Strong/Partial branches in Section 7 are the starting point for defining the ML/sensor scope in the next stage.